

The dimension ladder: from 0+0 to 1+3, and what the action constant does on each rung

Record of a discussion with the user, 2026-09-27, 02:50–03:15. The user proposed the frame: quantization is a consistency condition that Newton’s rework of the *Principia* missed, so any supposedly well-posed wave equation without a quantization carries a logical issue; the ladder of dimensions runs from 0 + 0 through Newton and quantum mechanics (1 + 0, “YM₀”) to 1 + 3; and the exponential $e^{-1/(2b_0\hbar g^2)}$ plays a distinct role. This note records the resulting statements with their status. The two layers of §1 and the table of §2 are the parts meant for the joint paper; §§3–4 are readings, labelled as such.

1. Quantization as a consistency condition: two layers

Layer 1: every classical wave equation needs a universal action constant to be thermodynamically consistent. Theorem U of the unit-and-indeterminacy note uses Maxwell’s field, but its mechanism is general. A classical field whose normal-mode frequencies are unbounded (every wave equation on a continuum) has, in equilibrium at temperature T , energy $k_B T$ per mode by equipartition, hence infinite energy density. A finite equilibrium, which the second law and every thermometer presuppose, needs the high modes to freeze out; that compares $k_B T$ with a frequency and so needs a constant with units of action. Two different fields in mutual equilibrium must share their occupation law, or a filter between them would run a perpetual motion of the second kind (Kirchhoff’s argument extended from colours to fields); so the constant is common to all fields. Einstein applied Planck’s constant to the vibrations of solids in exactly this way (Einstein 1907, metadata). Stated as a conjecture-level extension of Theorem U, to be proved in the same form: *any classical field with unbounded mode frequencies that admits a finite thermal equilibrium requires a universal action constant, and mutual equilibrium makes it the same for all fields.* A well-posed wave equation remains consistent as deterministic mathematics; as physics it is inconsistent once it can exchange energy with thermal matter. This is the kind of gap the programme attributes to Newton’s rework: the limits of the *Principia* are consistent, and the inconsistency appears when they are combined with thermodynamics and physical records.

Layer 2: the quantized theory must itself survive refinement. Once $\hbar$ is present, the per-cell parameter of the lattice refinement is $t = \hbar g_{\text{cl}}^2 a^{4-D}$ (zero-spacing note, §3):

- $D < 4$: $t \rightarrow 0$ and the refinement converges (superrenormalizable);
- $D = 4$: marginal; the theory survives only if its coupling runs to zero at short distances. Yang–Mills does (asymptotic freedom). Continuum ϕ_4^4 is trivial, a free field (Aizenman 1982; Fröhlich 1982; Aizenman and Duminil-Copin 2021; metadata), and QED has a Landau pole; on this reading both classical equations are effective theories with a cutoff;
- gravity: $t = \hbar G/(c^3 a^2) = (\ell_P/a)^2$ grows as $a \rightarrow 0$, the failure mode of the $D > 4$ row.

What “consistency” means in each layer. Layer 1 is a logical inconsistency: classical fields, the second law and finite energy are jointly contradictory without an action constant. Layer 2 is an existence condition on the continuum quantum theory, decided by estimates of the kind proved in the series/parallel and zero-spacing notes. Neither makes a classical PDE self-contradictory; both say it cannot be a complete physical theory by itself. Classical Yang–Mills in 3 + 1 Minkowski space exists globally (Eardley and Moncrief 1982, metadata), and classical ϕ_4^4 is well posed, although its continuum quantum theory is free: the large-quantum-number limit of that quantum theory is a free classical field. The implication “quantum theory exists $\Rightarrow$ its classical limit exists” holds in the sense of limits; the converse fails as mathematics. The Euclidean formulation makes one version exact: the measure $e^{-S/\hbar}$ is a classical statistical

field theory in D dimensions with $\hbar$ as temperature, so “quantum YM_4 exists” (Osterwalder–Schrader) and “classical statistical YM_4 at temperature $\hbar$ exists in the continuum” are one problem, while the classical deterministic PDE is a different object.

2. The ladder, in the user’s $1 + n$ labelling

Write $\lambda_D = \hbar g_{\text{cl}}^2$ with $[\lambda_D] = \text{length}^{D-4}$, $D = 1 + n$, and hold the classical coupling g_{cl} fixed as $\hbar \rightarrow 0$.

$1 + n$	Theory	Scale from $\hbar, c,$ g_{cl}	$\hbar$ -dependence	Status
$0 + 0$	single integral $\int dU e^{-S/\hbar}$	no time, no spectrum	series in $\hbar$ plus $e^{-c/\hbar}$	integrals
$1 + 0$	Newton, QM (“ YM_0 ”)	record floor on action; rotor $\hbar^2 C_2/(2I)$	$\hbar$ (action), $\hbar^2$ (rotor)	Planck paper; rotor exact
$1 + 1$	YM_1	$\hbar c \sqrt{\lambda_2}$; circle $\hbar c \lambda_2 L C_2/2$	$\hbar^{3/2}$; $\hbar^2 L$	exact; no local particle
$1 + 2$	YM_2	$C_3 \hbar c \lambda_3$	$\hbar^2$	C_3 open
$1 + 3$	YM_3	$C_4 \hbar c \Lambda$	$e^{-1/(2b_0 \hbar g_{\text{cl}}^2)}$	the Millennium problem

(The subscripts of the series/parallel and zero-spacing notes count spacetime dimension D ; here YM_n is $D = n + 1$.)

$0 + 0$. There is no time direction, so no Hamiltonian and no gap. What remains is one integral. It is the building block of every rung: the plaquette weight $k_t(U)$ is a $0 + 0$ theory, series moves (convolution) and parallel moves (products) assemble higher dimensions from it, and the elimination of one inserted Newtonian time (Proposition 2 of the refinement note) is the $0 + 0$ piece of mechanics. A zero-dimensional integral such as $\int d\phi e^{-(\phi^2/2 + g\phi^4)/\hbar}$ already has an asymptotic, non-Borel-summable series in $\hbar$ with ambiguities of order $e^{-c/\hbar}$. At large N the one-plaquette unitary matrix integral, the building block of two-dimensional lattice Yang–Mills, has a third-order phase transition (Gross and Witten 1980, metadata): non-analyticity from a zero-dimensional integral in a limit that plays the role of $a \rightarrow 0$.

$1 + 1$ on a circle is exactly a $1 + 0$ rotor. Eq. (10) of the refinement note, $E_R = \hbar c \lambda_2 L C_2(R)/2$, equals $\hbar^2 C_2(R)/(2I)$ with $I = 1/(c g_{\text{cl}}^2 L)$: a quantum rigid rotor on the gauge group whose moment of inertia is classical ($\hbar$ cancels from I). On the infinite line there is no particle; the string tension $\sigma = \hbar c \lambda_2 C_2/2$ between static charges goes like $\hbar^2$.

The exponents interpolate. The only energy formed from $\hbar, c$ and a fixed classical coupling is $E \sim \hbar c \lambda_D^{1/(4-D)}$, which goes like $\hbar^{(5-D)/(4-D)}$: $\hbar^{3/2}$ for $D = 2$, $\hbar^2$ for $D = 3$, and a diverging exponent as $D \rightarrow 4$. At $D = 4$ no power is available, and the running coupling turns the divergent power into $e^{-1/(2b_0 \hbar g_{\text{cl}}^2)}$, as dimensional regularization sees it ($(g^2)^{1/\varepsilon}$ with $\varepsilon = 4 - D$ becomes an exponential of $-1/g^2$). This is dimensional analysis; whether each rung has a positive gap is the open content (C_3 , and the Millennium problem).

Classical Yang–Mills has no gap in any dimension: its waves are massless and carry arbitrarily small energy. On every rung the large-quantum-number limit is the limit in which the gap disappears.

Refinement runs in opposite directions in mechanics and in field theory. For $D < 4$, $\hbar \rightarrow 0$ and $a \rightarrow 0$ push the same per-plaquette parameter $t = \hbar g_{\text{cl}}^2 a^{4-D}$ to zero: per plaquette, “more classical” and “finer lattice” are one direction, while the global scale where a gap lives stays quantum. In mechanics, refining the time cells makes each cell more quantum: the classical polygon action per cell shrinks like τ^3 , while the fluctuation action per cell stays of order $\hbar$.

3. A third route to a universal action: gravity with a least length

Classical gravitation with finite light speed has no mass-independent action (Gm^2/c depends on the mass). A fixed length ℓ supplies one: $c^3\ell^2/G$. So (G, c, ℓ) is a third route to a universal action, beside radiation thermodynamics (Theorem U) and the Stoney unit k_e/c , and the most Newtonian of the three: G is Newton's constant, c is Rømer's (1676), and the missing ingredient is exactly the least magnitude that the scholium closing Book I, Section I refuses by citing Euclid X. The value needs $\ell = \ell_P \approx 1.6 \times 10^{-35}$ m; Newton's interval of fits (1/89000 inch) gives $c^3\Lambda^2/G \sim 10^{22}$ J s, off by about 56 orders of magnitude. The route is logically Newtonian; its value is outside Newton-age data.

4. The exponential ladder (a reading)

The factor $e^{-1/(2b_0\hbar g^2)}$ has four properties worth keeping together.

- It vanishes to all orders in $\hbar$, so no finite order of the loop expansion sees the gap.
- In four dimensions the running converts it into a power of length, $e^{-c/g^2(a)} = (a\Lambda)^{2b_0c}$: dimensional transmutation. This is why the large-field criterion of the zero-spacing note is a power-law condition, $2b_0c > 4$, and why small instantons go like $(a\Lambda)^{11N/3}$.
- Every scale of pure YM₄ then reads as a suppression with an “action”: $\Lambda^k \propto e^{-k/(2b_0g^2)}$. The gap behaves like a configuration of action $1/(2b_0)$ in units of g^{-2} , which is $24\pi^2/33 \approx 7.2$ for $SU(3)$, against $8\pi^2 \approx 79$ for an instanton; the dislocation threshold is $k = D = 4$ times it. This is a reading, not a result.
- It lies where perturbation theory is ambiguous: the series in g^2 is not Borel summable, with ambiguities of order $e^{-\text{const}/g^2}$, powers of Λ ('t Hooft's renormalons). The gap is in the class of quantities the perturbative expansion cannot decide, which is why non-perturbative estimates of the H1 type are unavoidable.

The same exponential runs through the programme: Theorem 5's $e^{-\pi^2/(8\lambda_3 a)}$ in the series/parallel note, the instanton $e^{-8\pi^2/g^2}$, Wien's tail $e^{-h\nu/k_B T}$ (where quantization enters Theorem U, through the lumps $h\nu$), and on Newton's side the action as a phase $e^{iS/\hbar}$ or, in Euclidean form, $e^{-S/\hbar}$. In each case the exponential is where an action constant meets a threshold. The paper could organize the comparison by what each exponential suppresses and what survives refinement once the exponent is converted into a scale.

5. Consequence for STATE

No change to the queue. The table of §2 is proposed as the spine of the joint paper's comparison section, and Layer 1 of §1 as a theorem to prove in the form of Theorem U for a general wave field (with the cross-field universality). The gravity route of §3 belongs with the dimensional note when that note is next revised.